

PRIME-LOCAL FABEL FIBRES AND COMPLETE FIXED-COFACTOR CHANNEL RETURNS

THE CLANKERS

ABSTRACT. The reciprocal cubic of every marked first-half Erdős–Straus state is a literal target of the four-dimensional polynomial used in the ES–Fabel and Split-Zero conductor comparison. We determine its complete local fibre algebra at the original prime. Exterior states yield two ramified quadratic factors; middle states yield two unramified quadratic factors. Both have exactly three rational local inverse points, with all seven geometric affine branches and integral specializations accounted for. Separately, negative-four-square shape coordinates give original same-prime middle states on an exact divisor domain. The complete divisor fibre is then proved: for fixed source cofactor h , it is in bijection with every original middle state whose normalized cofactor is $Q = h$, and it strictly extends the two selected square words. All arithmetic coordinates, fibres and ordered denominator returns are explicit. Neither conclusion proves universal Erdős–Straus occupancy.

1. RESULT AND SCOPE

This is a continuation of the unresolved Turn 7 existence task. The original four-dimensional Fabel polynomial in the Zeta workbench admits a literal arithmetic specialization at the reciprocal cubic of an Erdős–Straus state. Its two channels have different local fibre algebras at the original prime: the exterior channel supplies ramified quadratic branches, whereas the middle channel supplies unramified quadratic branches. Both have exactly three rational local inverse points. The derivative, the full seven-point affine fibre, its integral specialization and the missing projective sign are retained.

A separate constructive continuation extends the earlier bounded-shape E-to-M returns to arbitrary negative-four-square shape coordinates on their exact integer domain. In particular the entire lines $\alpha = -4$ and $\beta = -4$ have original middle returns. Neither result produces the initial state at an arbitrary prime. No universal ES assertion, completed Turn 7 existence milestone, historical-priority determination or whole-programme analytic identification is claimed.

The polynomial and global inverse formulas are antecedents from the Zeta source [2, GF1–11, FC32–35]. Its four-dimensional map is the ES reader’s extension, not an additional preimage attributed to the original three-dimensional Alpöge–Fabel example discussed by Tao [6]. The classical Type-I/II coordinates are credited to [1, §2].

2. ORIGINAL ARITHMETIC AND THE COMMON RECIPROCAL CUBIC

Fix a prime $p \equiv 1 \pmod{4}$. Retain an original first-half source

$$p/4 < a < p/2, \quad R = 4a - p, \quad u \mid a^2, \quad E : R \mid 4u + 1, \quad M : R \mid u + a.$$

For $a = \prod_l l^{e_l}$ retain $\beta_l = v_l(u) - e_l \in [-e_l, e_l]$. With $g = \gcd(a, u)$ the original normalization is

$$h = g^2/u, \quad r = u/g, \quad s = a/g, \quad a = hrs, \quad u = hr^2, \quad \gcd(r, s) = 1.$$

For E let $\kappa = (pr + s)/R$ and return $(x_1, x_2, x_3) = (a, h s \kappa, p h r \kappa)$. For M let $\lambda = (r + s)/R$ and return $(x_1, x_2, x_3) = (a, p h s \lambda, p h r \lambda)$. Multiplication verifies the reciprocal identity. The ordered inverses are $u = a^2/(R x_2 - p a)$ and $u = p a^2/(R x_2 - p a)$, respectively. The M complement $u \mapsto a^2/u$ exchanges r, s and the last two denominators; its orientation is retained rather than suppressed.

Every normalization factor is a p -unit. For E use $pD + 1 = 4hr\kappa$, with $D = (4u + 1)/R = (r + \kappa)/s$; in particular κ need not be smaller than p . For M, $h, r, s < p$ and $\lambda = (r + s)/R < p$.

The three denominators are distinct. In E, $\kappa > r$ and $pr > s$; in M, $r = s$ would force $r = s = 1$ and $R \mid 2$. Their first denominator is below $p/2$ and the other two are above $p/2$.

Lemma 2.1 (Original character distinction). *For every original divisor $d \mid a^2$, $(d/R) = (d/p)$, with Jacobi symbol on the left. Every E state has $(h/p) = -1$. Every M state has $(rs/p) = -1$.*

Proof. For odd $l \mid a$, quadratic reciprocity and $R \equiv -p \pmod{l}$ give $(l/R) = (-1/l)(R/l) = (p/l) = (l/p)$. If $2 \mid a$, then $R \equiv -p \pmod{8}$ and the supplementary law gives $(2/R) = (2/p)$. Multiply over the actual exponents. At E, $u \equiv -1/4 \pmod{R}$ and $R \equiv 3 \pmod{4}$, so $(u/p) = -1$. Since $u = hr^2$, this proves the first assertion. At M, $r \equiv -s \pmod{R}$ and all the factors are units, so $(r/p) = (r/R) = -(s/R) = -(s/p)$. This proves $(rs/p) = -1$. These are the classical laws in [4, §6.4], applied to the retained original factors. $\square$

Define the labelled reciprocal roots and polynomial

$$t_i = p/x_i, \quad f(T) = \prod_{i=1}^3 (T - t_i) = T^3 - 4T^2 + \sigma_2 T - \sigma_3. \quad (1)$$

Then $2 < t_1 < 4$ and $0 < t_2, t_3 < 2$. The target for the *existing* Fabel map is

$$\mathbf{U} = (0, -4, \sigma_2, -\sigma_3), \quad H_{\mathbf{U}}(U, V) = V \prod_i (U - t_i V). \quad (2)$$

The leading zero and the coefficient of $U^3 V$, namely one, are fixed, not rescaled away. The inverse on the actual arithmetic domain is $x_i = p/t_i$, with its positivity and integrality tests and the original E/M divisor inverse. The coefficient target alone forgets the ordering; p and the labels are kept.

The preceding shape curve has different coordinates: $\alpha = v - R, \beta = v - D, X = v, Y = 2a$, with the additional fixed-prime line $2Y = p + X - \alpha$. Its inverse is $a = Y/2, R = X - \alpha, D = X - \beta, u = (RD - 1)/4$. Thus the connection to (2) is the explicit composition through this original state. No identification of two unmarked cubic curves is asserted.

3. THE LITERAL FABEL MAP AND EVERY INVERSE BRANCH

Use ξ for its first source coordinate, distinct from the original a . Fix $i^2 = -1$ and put $P = (P_0, P_2, P_3, P_4)$, where

$$\begin{aligned} P_0 &= \xi^3 z + 2\xi^2 y - i\xi, \\ P_2 &= -\xi^3 y^2 z - 2i\xi^2 yz + \xi^2 w - 2\xi^2 y^3 - 10i\xi y^2 + 3\xi z + y, \\ P_3 &= 2\xi^3 y^3 z + 6i\xi^2 y^2 z + 2\xi^2 wy + 4\xi^2 y^4 + 2i\xi w - 4i\xi y^3 - 2\xi yz + 2iz + 7y^2, \\ P_4 &= 2\xi^3 y^4 z + 8i\xi^2 y^3 z + \xi^2 wy^2 + 4\xi^2 y^5 + 2i\xi wy + 7i\xi y^4 - 10\xi y^2 z - 4iyz - w - 3y^3. \end{aligned} \quad (3)$$

Its factor coordinates are

$$\begin{aligned} b &= i + \xi y, & c &= -i + 2\xi y + \xi^2 z, \\ d &= -iy - \xi(iz + 2y^2) - \xi^2 yz, & e &= 2z - 7iy^2 + \xi w, \\ f_0 &= iw + 3iy^3 - 4yz + \xi(6iy^2 z + wy + 4y^4) + 2\xi^2 y^3 z. \end{aligned}$$

Writing $L = \xi U + bV$ and $C = cU^3 + dU^2V + eUV^2 + f_0V^3$, substitution gives

$$\xi d + bc = 1, \quad C(-b, \xi) = 1, \quad LC = P_0 U^4 + U^3 V + P_2 U^2 V^2 + P_3 U V^3 + P_4 V^4. \quad (4)$$

For a simple finite root t_i of (1), let $d_i = f'(t_i) = (t_i - t_j)(t_i - t_k)$. The full inverse is

$$\boxed{\xi^2 = d_i^{-1}, \quad y = -t_i - i/\xi, \quad z = 2t_i/\xi + 3i/\xi^2, \quad w = 7it_i^2/\xi + (-4 - 17t_i)/\xi^2 - 13i/\xi^3.} \quad (5)$$

Both signs of ξ are retained. The finite marked root is recovered by $t_i = -b/\xi$, and $C(-b, \xi) = \xi^2 f'(t_i) = 1$. Coefficient comparison in (4) determines the quotient factor uniquely; the factor-coordinate inverse $y = (b - i)/\xi, z = (c + i - 2\xi y)/\xi^2, w = (e - 2z + 7iy^2)/\xi$ then proves (5) and its exhaustiveness.

At $\xi = 0$ the single chart inverse is

$$q_\infty = (0, -4, i(112 - \sigma_2)/2, -704 + 8\sigma_2 + \sigma_3). \quad (6)$$

It has $b = i, c = -i$. The other signed projective infinity factor has $b = -i$ and is outside this affine chart. Thus (2) has *seven*, not eight, geometric affine preimages: six finite ones and (6). The missing infinity sign is different from any arithmetic field or integrality obstruction.

For completeness the constant Jacobian statement used here is verified as follows. On $\xi \neq 0$ use (ξ, t, A, B) , with $A = P_0, B = P_2$. The target equations are

$$P_3 = \xi^{-2} - 4At^3 - 3t^2 - 2Bt, \quad P_4 = 3At^4 + 2t^3 + Bt^2 - t\xi^{-2}.$$

The full source inverse has determinant $-\xi^{-5}$ in those coordinates; the target determinant is $2\xi^{-5}$. Their ratio is -2 . Polynomial continuation proves $\det DP = -2$ everywhere. These are the antecedent GF10–11 calculations [2], with their constant and sign retained; the verifier independently expands the original derivative at every recorded inverse point.

4. THE COMPLETE LOCAL ALGEBRA DISTINGUISHES E FROM M

Fix either root $i_p \in \mathbb{Q}_p$ of -1 . Its existence follows by lifting a root modulo $p \equiv 1 \pmod{4}$. Normalize $v_p(p) = 1$ on all extensions. The reduced coordinate algebra of the actual fibre is

$$\mathcal{A}_{p,x} = \mathbb{Q}_p \times \prod_{i=1}^3 \mathbb{Q}_p[\xi_i]/(\xi_i^2 - d_i^{-1}). \quad (7)$$

The first factor is the single infinity chart. Formula (5) and its inverse identify the other factors; the derivative is nonzero, so there are no unrecorded nilpotents. This is the coordinate algebra of the fixed target, not a newly chosen seven-point model.

Theorem 4.1 (Prime-local arithmetic fibre). *For every original first-half ES state at a prime $p \equiv 1 \pmod{4}$, $P^{-1}(\mathbf{U})$ has exactly three $\mathbb{Q}_p$ -rational points. More precisely,*

$$\begin{aligned} E : \quad \mathcal{A}_{p,x} &\simeq \mathbb{Q}_p^3 \times K_{\text{ram}} \times K_{\text{ram}}, \\ M : \quad \mathcal{A}_{p,x} &\simeq \mathbb{Q}_p^3 \times K_{\text{unr}} \times K_{\text{unr}}, \end{aligned} \quad (8)$$

where $K_{\text{ram}}/\mathbb{Q}_p$ is a ramified quadratic extension and $K_{\text{unr}}/\mathbb{Q}_p$ is the unramified quadratic extension. The repeated field factors retain their distinct root labels. There is no fully split seven-point local fibre in the original arithmetic domain.

Proof. For E put $H = hr\kappa$. Then

$$\begin{aligned} (t_1, t_2, t_3) &= H^{-1}(p\kappa, pr, s), \\ (d_1, d_2, d_3) &= H^{-2}(p(\kappa - r)(p\kappa - s), -p(\kappa - r)(pr - s), (p\kappa - s)(pr - s)). \end{aligned} \quad (9)$$

All factors h, r, s, κ are p -units. Also $p \nmid \kappa - r$: otherwise $4hr\kappa \equiv 1 \pmod{p}$ would make $4hr^2 \equiv 1$, contrary to Lemma 2.1. Hence

$$(v_p(d_1), v_p(d_2), v_p(d_3)) = (1, 1, 0), \quad d_1/d_2 \equiv -1 \pmod{p}, \quad d_3 \equiv 16 \pmod{p}. \quad (10)$$

A unit congruent to a nonzero square modulo the odd prime p is a square in $\mathbb{Q}_p$ by Hensel lifting. Thus d_1/d_2 is a square and the first two quadratic factors are copies of one ramified field (their defining square classes have odd valuation). The last quadratic splits into two rational points. Together with infinity this proves the E case.

For M put $H = hrs\lambda$. Then

$$(t_1, t_2, t_3) = H^{-1}(p\lambda, r, s).$$

The numerator differences are units: $r, s < p$, $r \neq s$, and $p \nmid r - s$. Modulo p , the derivative classes, up to the common square H^{-2} , are

$$rs, \quad r(r - s), \quad s(s - r). \quad (11)$$

The first is a nonsquare by Lemma 2.1. Their product is $-r^2s^2(r - s)^2$, a square since -1 is a square modulo p . Thus exactly one of the three classes is a square, and exactly two are nonsquares.

The two nonsquare units have square ratio and define the same unramified quadratic extension. The remaining quadratic gives two rational points. This proves (8) and the rational-point count. $\square$

The local facts used in this proof can be made constructive. A root b_n of a unit z modulo p^n has the unique lift

$$b_{n+1} = b_n + p^n t, \quad t \equiv -\frac{b_n^2 - z}{p^n} (2b_n)^{-1} \pmod{p}.$$

Use the compatible rational residue of z to compute the integer quotient. A nonsquare residue has irreducible separable quadratic reduction, giving the unramified degree-two extension; a class p times a unit has square root of valuation $1/2$, giving ramification index two. These are the standard local field facts in [5]; the displayed lift is also checked directly.

An algebra isomorphism over $\mathbb{Q}_p$ must permute the primitive field factors. It cannot change their ramification indices. Therefore no such isomorphism carries the E fibre algebra to the M fibre algebra at the fixed prime. Equality of the rational-point counts (both three) does not give that isomorphism. This restricts an *unchanged fibre algebra*; it does not prohibit an arithmetic map to a different target, constructed below.

5. INTEGRAL SPECIALIZATION AND THE NONPROPER BOUNDARY

For E, the exact numerator formula further gives

$$v_p(\sigma_2) = 1, \quad v_p(\sigma_3) = 2, \quad v_p(\text{Disc } f) = 2, \quad \bar{f}(T) = T^2(T - 4). \quad (12)$$

Indeed $\sigma_2 = p\{p\kappa r + s(\kappa + r)\}/H^2$ and $\kappa + r = sD$ is a unit. If $p \mid D$, then $RD = 4u + 1$ would make the nonresidue u equal to the square $-1/4$ modulo p , impossible. The product gives σ_3 and the three differences give the discriminant.

For each of the four signed branches over t_1, t_2 , equation (5) has the exact valuation vector

$$(v_p(\xi), v_p(y), v_p(z), v_p(w)) = (-1/2, 1/2, 1, 1). \quad (13)$$

The least terms are respectively ξ , $-i/\xi$, $3i/\xi^2$ and $-4/\xi^2$. The coefficient 3 is a unit since $p \equiv 1 \pmod{4}$. There is no equal-valuation cancellation in these minima. These are genuine generic-field points but are not integral and have no affine specialization over the valuation ring. The two branches at t_3 and the infinity branch are integral.

Over an algebraic closure of the residue field, the complete E special fibre has precisely the following three points:

$$(0, -4, 56i_p, -704), \quad (1/4, -4 - 4i_p, 32 + 48i_p, -1152 - 384i_p), \\ (-1/4, -4 + 4i_p, -32 + 48i_p, -1152 + 384i_p). \quad (14)$$

The double root zero has no resultant-one factor scaling; the simple root four has derivative sixteen. Thus the reduction from seven geometric generic points to three special points is a loss at the affine boundary, not a vanishing source Jacobian. The determinant -2 remains a unit.

For M, the three roots have the distinct reductions $0, r/H, s/H$, and every derivative value is a unit. The derivative residues themselves need not be pairwise distinct: at the valid state

$$p = 5, \quad (a, u, R, h, r, s, \lambda, H) = (2, 1, 3, 1, 1, 2, 1, 2),$$

the reduced roots are $(0, 3, 1)$ whereas the derivative residues are $(3, 1, 3)$. Every geometric inverse in (5) is integral over an unramified extension. Its geometric special fibre still has seven points; precisely three are rational over $\mathbb{F}_p$. Thus counting only rational special points also fails to distinguish the two channels.

The exact relation to the original conductor. The Zeta source constructs fixed invertible complex linear maps Φ, Ψ with $T_A \Phi = \Psi$ and transported polynomials $\Phi P \Phi^{-1}, \Psi P \Psi^{-1}$ [2, FC32–35]. Consequently (5)–(6) transport point by point through that exact complex map, with its retained original Gram G and norm $q^* G q$. No Euclidean replacement, new coefficient fit or linear-conductor kernel is asserted. The p -adic statements here concern the original polynomial over $\mathbb{Q}(i)$ after the specified embedding in $\mathbb{Q}_p$. Complex period entries in Ψ are not thereby declared to have a $\mathbb{Q}_p$ -model. Such a descent would be an additional theorem.

A nonempty polynomial fibre is not an arithmetic solution. The target $(0, -4, 0, 1)$ has the exact infinity inverse $(0, -4, 56i, -705)$. Its cubic $T^3 - 4T^2 + 1$ has no rational root: the rational-root theorem leaves only ± 1 , neither a root. Thus it cannot be the reciprocal cubic of three positive integer denominators at any rational p . The inverse formula alone does not supply the initial arithmetic target.

6. THE EXACT INTEGRAL ROOT ORDER AND ITS OBLIGATORY DENOMINATOR

This is a further arithmetic distinction between the coefficient target and its split roots. Let $\mathcal{O}_f = \mathbb{Z}_p[T]/(f)$, observed by evaluation at the three original labelled roots. All are integral, and f is monic.

Theorem 6.1 (Root-order defect and exact interpolation loss). *For an original E state,*

$$\boxed{\mathcal{O}_E \simeq \{(z_1, z_2, z_3) \in \mathbb{Z}_p^3 : z_1 \equiv z_2 \pmod{p}\}}. \quad (15a)$$

Its index in $\mathbb{Z}_p^3$ is p and its conductor ideal in that ring is $p\mathbb{Z}_p \times p\mathbb{Z}_p \times \mathbb{Z}_p$. For M , evaluation gives $\mathcal{O}_M \simeq \mathbb{Z}_p^3$.

Given any original E and M state at the same prime, let x_i and y_i be their ordered reciprocal roots. There is a unique polynomial F of degree at most two taking x_i to y_i , and a unique G of degree at most two taking y_i to x_i . Their exact formulas are

$$F(T) = \sum_i y_i \frac{(T - x_j)(T - x_k)}{(x_i - x_j)(x_i - x_k)}, \quad G(T) = \sum_i x_i \frac{(T - y_j)(T - y_k)}{(y_i - y_j)(y_i - y_k)}. \quad (15b)$$

Writing coefficients in increasing degree, F has exact p -valuations $(0, -1, -1)$, while $G \in \mathbb{Z}_p[T]$. Moreover $pF \pmod{p}$ is a nonzero scalar multiple of $T(T-4)$. The compositions satisfy $G(F(T)) = T \pmod{f_E}$ and $F(G(T)) = T \pmod{f_M}$.

Proof. For E , $x_1 - x_2$ has valuation one; both differences from x_3 are units. A polynomial over $\mathbb{Z}_p$ therefore has its first two values congruent modulo p . Conversely, for a tuple with that congruence, interpolation at x_1, x_2 is integral since its divided difference is integral. Its error at x_3 can be corrected by an integral multiple of $(T - x_1)(T - x_2)$, whose value there is a unit. This proves (15a). The quotient is exactly $\mathbb{F}_p$ by $z \mapsto z_1 - z_2 \pmod{p}$. Multiplication by an arbitrary tuple preserves the congruence exactly when its first two coordinates are divisible by p , which proves the conductor. All three differences for M are units, giving its integral interpolation inverse.

Formula (15b) is the original labelled interpolation. The E derivative valuations imply $pF \in \mathbb{Z}_p[T]$. It cannot be integral before that multiplication, since it takes two congruent arguments x_1, x_2 to the incongruent values y_1, y_2 . In fact modulo p , only its second Lagrange summand survives after multiplication by p : y_1 is divisible by p , while the third denominator is a unit. Therefore

$$pF(T) \equiv (py_2/d_2)T(T-4) \pmod{p},$$

with a nonzero coefficient. This proves valuation -1 for the linear and quadratic coefficients. The constant term is a unit: its second summand has valuation zero, and the first and third have positive valuation. The reverse interpolation has only unit denominators and integral numerators. Equality of each composition with T at all three distinct labelled roots proves the two polynomial congruences. $\square$

The exact sequence

$$0 \longrightarrow \mathcal{O}_E \longrightarrow \mathbb{Z}_p^3 \xrightarrow{z \mapsto z_1 - z_2 \pmod{p}} \mathbb{F}_p \longrightarrow 0$$

has no $\mathbb{Z}_p$ -linear right inverse, because the middle module is torsion-free. Tensoring with $\mathbb{Q}_p$ removes this index- p quotient; that operation loses original integral information. This is the root-order conductor, explicitly distinct from the Zeta workbench's weighted linear conductor T_A . We do not identify their source modules.

Thus there is a fully explicit rational root-level map between the two cubics, with a sharp integral obstruction and an integral reverse map. The lack of an isomorphism between their seven-point local fibre algebras is not described as unrelatedness. It is the additional quadratic-factor ramification in (8), which this root interpolation alone does not preserve.

For a small complete illustration, at $p = 5$ take the original E denominators $(2, 4, 20)$ and M denominators $(2, 10, 5)$. Their reciprocal root triples are $(5/2, 5/4, 1/4)$ and $(5/2, 1/2, 1)$. The exact maps are

$$\boxed{F(T) = \frac{14}{15}T^2 - \frac{19}{10}T + \frac{17}{12}, \quad G(T) = \frac{7}{4}T^2 - \frac{37}{8}T + \frac{25}{8}.}$$

The first requires exactly one power of five in its local denominator; $5F \equiv 3T(T - 4) \pmod{5}$. The reverse is integral at five. Both ordered root triples and their original denominator inverses are retained in the certificate.

7. AN UNBOUNDED FAMILY OF EXACT SAME-PRIME CHANNEL RETURNS

Continue with the original E state, and set $\alpha = hs^2 - R$, $\beta = hs^2 - D$. The earlier shape identities give

$$h \mid \alpha\beta - 1, \quad p \equiv \alpha \pmod{h}, \quad p\beta \equiv 1 \pmod{h}. \quad (15)$$

Theorem 7.1 (Negative-four-square channel transfer). *Suppose one marked coordinate is $-4c^2$, with $c > 0$ integral. Then $h \equiv 3 \pmod{4}$ and $\gcd(h, c) = 1$. Put $j = (h + 1)/4$. If $c \mid j$, set*

$$Q = h, \quad U = \begin{cases} c^2, & \alpha = -4c^2, \\ (j/c)^2, & \beta = -4c^2, \end{cases} \quad R_M = (p + 4U)/Q, \quad a_M = (p + R_M)/4. \quad (16)$$

Then $U \mid j^2$, $Q \mid p + 4U$, $0 < R_M < p$, and (16) is an original M state. With $g_M = \gcd(j, U)$, its complete normalization is

$$\boxed{h_M = g_M^2/U, \quad r_M = U/g_M, \quad \lambda_M = j/g_M, \quad s_M = R_M\lambda_M - r_M.} \quad (17)$$

The new h_M is a square. The ordered return is $(a_M, ph_M s_M \lambda_M, ph_M r_M \lambda_M)$. Every existing original E state on either shape line $\alpha = -4$ or $\beta = -4$ is covered, without a bound on its other coordinate. This does not assert that either line contains infinitely many source states, or that every prime supplies a state on one of the lines.

Proof. In the alpha branch, $R = hs^2 + 4c^2$ and $R \equiv 3 \pmod{4}$; in the beta branch, $D = hs^2 + 4c^2$ and $RD = 4hr^2 + 1$ with $R \equiv 3 \pmod{4}$, so again $D \equiv 3 \pmod{4}$. In either case s is odd and $h \equiv 3 \pmod{4}$. Equation (15) then gives $h \mid \alpha\beta - 1$, which makes h coprime to c . In the α case the required congruence is immediate. In the β case, $p \equiv -(4c^2)^{-1} \pmod{h}$ and $4j \equiv 1 \pmod{h}$, so $p + 4(j/c)^2 \equiv 0$. Availability follows from $c \mid j$ in both cases.

For α , the original inequality $R = hs^2 + 4c^2 < p$ implies $4U < p - h < (h - 1)p$. For β , use $4U \leq 4j^2 = (h + 1)^2/4$ and $p > 2h$; for $h \geq 3$, $(h + 1)^2 < 8h(h - 1)$, so again $4U < (h - 1)p$. Thus $R_M < p$. Its residue is $3 \pmod{4}$, proving the original first-half range.

The valuation normalization of $U \mid j^2$ gives $h_M r_M \lambda_M = j$, $h_M r_M^2 = U$, and $\gcd(r_M, \lambda_M) = 1$. Also $a_M = R_M j - U = h_M r_M s_M > 0$. Since $r_M \leq j < p$, the equation $QR_M = p + 4U$ proves $\gcd(r_M, R_M) = 1$, hence $\gcd(r_M, s_M) = 1$. This is therefore precisely the original M normalization.

For an explicit square-grade formula write $j = cm$, $d = \gcd(c, m)$. Then

$$\begin{array}{c|ccc} & h_M & r_M & \lambda_M \\ \hline \alpha & d^2 & c/d & m/d \\ \beta & d^2 & m/d & c/d. \end{array}$$

This also proves that the original shared budget has not been used twice. $\square$

The condition $c \mid j$ is the exact availability domain of the two selected cofactor words: $c^2 \mid j^2$ if and only if $c \mid j$, and j/c must be integral. It is not a hypothesis inserted into a universal existence theorem. Its failure can leave other words available, or the entire selected cofactor box empty. The square-divisor criterion in (16) is classical Type-II arithmetic; what is proved here is its supply from these actual exterior shapes.

7.1. Complete inverse fibres and coefficient collisions. The target retains its cofactor $Q = 4h_M r_M \lambda_M - 1$, so it recovers the original h . In the α branch it recovers $c = \sqrt{U}$; in the β branch $c = j/\sqrt{U}$. The square roots here are positive integers.

For fixed (p, h, c, α) , put $N = (p + 4c^2)/h$. Every original preimage is obtained by $s \mid N$, $k = N/s$, $s+k \equiv 0 \pmod{4}$, $r = (s+k)/4$, $a = hrs$, $u = hr^2$, $R_s = hs^2 + 4c^2$, subject to

$$\gcd(r, s) = 1, \quad R_s < p, \quad R_s \mid 4hr^2 + 1. \quad (17a)$$

The last divisibility is the original E gate; equivalently $R_s \mid pr + s$. Conversely, these tests give $4hrs - p = R_s$, $u \mid a^2$, and the exact normalized E state. Indeed, if $D = (4hr^2 + 1)/R_s$, then

$$pr + s = s(4hr^2 + 1) - rR_s = R_s(sD - r),$$

so $\kappa = sD - r$ is integral and $hs^2 - R_s = -4c^2$. Hence the displayed list, including (17a), is the full alpha fibre. The gate cannot be dropped: at $p = 5209, h = 95, c = 2$, the candidate $s = 5, k = 11, r = 4$ passes the displayed divisor, parity, gcd and range tests but has $R_s = 2391 \nmid 6081 = 4hr^2 + 1$.

For fixed (p, h, c, β) , put $F = (4c^2 p + 1)/h$. For $1 \leq s \leq \lfloor (p-1)/(2h) \rfloor$, set

$$D = hs^2 + 4c^2, \quad \Delta_s = s^2(D^2 - p) - F.$$

Keep exactly the positive squares $k^2 = \Delta_s$ for which $sD - k$ is positive and even, and set $r = (sD - k)/2$, $\kappa = sD - r$. The original coprimality and range tests complete the fibre. This comes from $4hr(sD - r) = pD + 1$ and $k = \kappa - r > 0$, so both directions are proved.

Within each marked branch, c is recovered. Equality of the full selected targets first forces the same retained cofactor $Q = 4h_M r_M \lambda_M - 1 = h$. For this common h and $j = (h+1)/4$, an α branch with c_1 and a β branch with c_2 select the same U exactly when $c_1 c_2 = j$. Different original sources in any one fibre remain separate integer coefficients. Pushing them to one target sums those coefficients; the kernel is generated by their same-target differences. No injectivity or one-to-one state count is claimed after those records are discarded.

7.2. An actual transition, with local ramification changed. At the hard prime $p = 5209$, the original E state is

$$(a, R, u, h, r, s, \kappa) = (1330, 111, 18620, 95, 14, 1, 657).$$

Its shape is $(-16, -576)$, so the two selected branches have $c_1 = 2, c_2 = 12$ and $j = 24$. Both supply the *same* target $U = 4$. Formula (17) gives

$$(a_M, R_M, u_M, h_M, r_M, s_M, \lambda_M) = (1316, 55, 4, 4, 1, 329, 6),$$

$$\boxed{\frac{4}{5209} = \frac{1}{1316} + \frac{1}{41130264} + \frac{1}{125016}}. \quad (18)$$

The original E box uses primes 2, 5, 7, 19. Its M target uses 2, 7, 47; the factor 47 is part of the reconstructed integer, not an available original E factor silently adjoined. At this same prime the original Fabel fibre changes from ramified to unramified by Theorem 4.1, because the target cubic has changed through an actual arithmetic return. No local algebra isomorphism between the unchanged fibres is being asserted.

A different hard example is

$$p = 4561, \quad (a, R, u, h, r, s, \kappa) = (1593, 1811, 43011, 59, 27, 1, 68), \quad \beta = -36.$$

Here $c = 3, j = 15$ returns

$$(a_M, R_M, u_M, h_M, r_M, s_M, \lambda_M) = (1160, 79, 25, 1, 5, 232, 3),$$

with denominators (1160, 3174456, 68415).

At the hard prime $p = 3049$, the original E state

$$(a, R, u, h, r, s, \kappa) = (806, 175, 20956, 31, 26, 1, 453)$$

has $\alpha = -144$, hence $c = 6, j = 8$. The selected $U = 36$ is unavailable in $j^2 = 64$. More strongly, the entire cofactor box has divisors 1, 2, 4, 8, 16, 32, 64, none satisfying $31 \mid 3049 + 4U$; the required residue is 5 (mod 31). A different original shell has the middle state

$$(a, R, u, h, r, s, \lambda) = (770, 31, 5, 5, 1, 154, 5).$$

This is an exact local obstruction, not an ES counterexample.

8. VERIFICATION, SOURCE DISTINCTIONS AND REMAINING TASK

The release verifier enumerates the complete first-half boxes for all primes $p \equiv 1 \pmod{4}$ through its declared bound, checks every labelled E/M state, its ordered identity, the local field signatures, and every present negative-square crossing. It evaluates (3) on all seven exact branches of representative fibres in the four-dimensional rational algebra generated by i and ξ ; it does not assume that this algebra is a field. It computes the original Jacobian by exact first-order arithmetic and checks the unit-root lifts modulo p^4 .

The separate program imports none of the main implementation or predecessor software. It enumerates primitive h, r, s to recover the full state lists, checks the factor-coordinate product (4), independently enumerates the inverse crossing fibres, and exhausts the original affine polynomial fibre over $\mathbb{F}_{17}$ at both special targets. Execution receipts distinguish these checks from a mathematical review or a Lean build.

The positive progress is an explicit same-prime return on unbounded shape lines, and an exact local arithmetic meaning for the inverse branches read in the Zeta workbench. The root-field signature does not prove that an ES reciprocal target exists, nor that every remaining exterior shape reaches a middle state. Turn 7's universal positive-coefficient obligation remains unresolved. We do not advance to Turn 8 under an assumed occupancy statement.

9. THE COMPLETE FIXED-COFACTOR MIDDLE FIBRE

The selected square words in Theorem 7.1 sit inside a larger finite fibre which can be computed without guessing a word.

Theorem 9.1 (Complete fixed- Q middle fibre). *Let*

$$(p, a, R, u, h, r, s, \kappa, D, \alpha, \beta)$$

be an original E state, in the notation of the preceding sections, and assume $h \equiv 3 \pmod{4}$. Put

$$j = \frac{h+1}{4}, \quad \mathcal{U}_{p,h} = \{U \in \mathbb{Z}_{>0} : U \mid j^2, h \mid p + 4U\}.$$

For $U \in \mathcal{U}_{p,h}$, define

$$\begin{aligned} R_U &= \frac{p+4U}{h}, & a_U &= \frac{p+R_U}{4}, & g_U &= \gcd(j, U), \\ h_U &= \frac{g_U^2}{U}, & r_U &= \frac{U}{g_U}, & \lambda_U &= \frac{j}{g_U}, & s_U &= R_U \lambda_U - r_U. \end{aligned} \tag{19}$$

Then

$$\boxed{(a_U, R_U, U, h_U, r_U, s_U, \lambda_U)}$$

is an original M state at the same prime. Its middle cofactor is exactly

$$Q_U = 4h_U r_U \lambda_U - 1 = h,$$

and its ordered reciprocal decomposition is

$$\boxed{(a_U, ph_U s_U \lambda_U, ph_U r_U \lambda_U)}. \quad (20)$$

The assignment $U \mapsto (a_U, R_U, U, h_U, r_U, s_U, \lambda_U)$ is a bijection from $\mathcal{U}_{p,h}$ onto all original M states at p whose normalized cofactor is $Q = h$.

Proof. Because $h \leq a < p/2$, $p > 2h$, and $h \geq 3$. From $U \mid j^2$,

$$4U \leq 4j^2 = \frac{(h+1)^2}{4} < 2h(h-1) < p(h-1).$$

Consequently $0 < R_U < p$. Since $hR_U = p + 4U \equiv 1 \pmod{4}$ and $h \equiv 3 \pmod{4}$, one has $R_U \equiv 3 \pmod{4}$. Thus a_U is integral and $p/4 < a_U < p/2$.

Prime by prime, $U \mid j^2$ implies $g_U^2/U \in \mathbb{Z}_{>0}$. The definitions give

$$h_U r_U = g_U, \quad h_U r_U^2 = U, \quad h_U r_U \lambda_U = j.$$

Moreover,

$$R_U j - U = \frac{R_U(h+1)}{4} - U = \frac{p + R_U}{4} = a_U.$$

It follows that $h_U r_U s_U = a_U$ and $r_U + s_U = R_U \lambda_U$. Since $\gcd(r_U, \lambda_U) = 1$, a common prime divisor of r_U, s_U would divide r_U, R_U , and hence $hR_U - 4U = p$. It cannot equal p , because $0 < R_U < p$. Thus $\gcd(r_U, s_U) = 1$. Finally

$$a_U + U = R_U j,$$

so $R_U \mid a_U + U$, and $U = h_U r_U^2 \mid a_U^2$. This proves the forward construction and $4h_U r_U \lambda_U - 1 = 4j - 1 = h$.

Conversely, take an original M state with cofactor $Q = h$, normalized as

$$a_M = h_M r_M s_M, \quad u_M = h_M r_M^2, \quad \lambda_M = \frac{r_M + s_M}{R_M}.$$

Then

$$j = \frac{h+1}{4} = h_M r_M \lambda_M, \quad \frac{j^2}{u_M} = h_M \lambda_M^2 \in \mathbb{Z}_{>0}.$$

Therefore $U = u_M \mid j^2$. Also

$$a_M + u_M = h_M r_M (r_M + s_M) = R_M j,$$

whence

$$p + 4U = 4(a_M + u_M) - R_M = R_M(4j - 1) = R_M h.$$

The primitive normalization gives $\gcd(j, U) = h_M r_M$, and (19) recovers h_M, r_M, λ_M, s_M exactly. Distinct U 's remain distinct as original middle divisors. This proves bijectivity. $\square$

Corollary 9.2 (Exact negative-square divisor slices). *If an original E state has $\alpha = -4c^2$ or $\beta = -4c^2$, then*

$$h \equiv 3 \pmod{4}, \quad \gcd(h, c) = 1,$$

and, with $j = (h+1)/4$,

$$\alpha = -4c^2 : \quad \mathcal{U}_{p,h} = \{U \mid j^2 : U \equiv c^2 \pmod{h}\}, \quad (21)$$

$$\beta = -4c^2 : \quad \mathcal{U}_{p,h} = \{U \mid j^2 : c^2 U \equiv j^2 \pmod{h}\}. \quad (22)$$

The words $U = c^2$ in alpha and $U = (j/c)^2$ in beta are available exactly when $c \mid j$. That divisibility is not necessary for the full fixed- Q fibre to be nonempty.

Proof. The congruence and coprimality assertions were proved in Theorem 7.1. In the alpha case, $p \equiv -4c^2 \pmod{h}$, so $h \mid p + 4U$ is equivalent to (21). In the beta case, $-4c^2 p \equiv 1 \pmod{h}$. Multiplication of $p + 4U \equiv 0 \pmod{h}$ by $4c^2$, together with $4j \equiv 1 \pmod{h}$, gives (22), and every step is reversible because $\gcd(4c^2, h) = 1$. $\square$

Define the finite residue set

$$\Delta_h = \{U \bmod h : U \mid j^2\} = \left\{ \prod_{\ell^e \parallel j} \ell^{f_\ell} \bmod h : 0 \leq f_\ell \leq 2e \right\}.$$

The exact emptiness criterion is

$$\boxed{\mathcal{U}_{p,h} = \emptyset \iff -p4^{-1} \pmod{h} \notin \Delta_h.} \quad (23)$$

Thus the fixed- Q question is a complete finite divisor-residue fibre, not merely the selected square test. Its other U 's can be nonsquares, so the square-grade conclusion of Theorem 7.1 remains specific to its selected words.

9.1. Composition with the diagonal and radius-eight returns. For a hard prime, let $\mathcal{E}_p$ be the complete original E-state set,

$$\mathcal{D}_p = \{e \in \mathcal{E}_p : R = D\},$$

$$\mathcal{F}_p = \{e \in \mathcal{E}_p : h \equiv 3 \pmod{4}, \mathcal{U}_{p,h} \neq \emptyset\}.$$

After applying the proved diagonal return and every fixed- Q return, the exact residual source is

$$\boxed{\mathcal{E}_p^{\text{res}} = \{e \in \mathcal{E}_p : R \neq D, [h \not\equiv 3 \pmod{4} \text{ or } \mathcal{U}_{p,h} = \emptyset]\}.} \quad (24)$$

Every state in (24) has

$$\boxed{\max(|\alpha|, |\beta|) \geq 9.} \quad (25)$$

Indeed, radius at most seven is the diagonal profile $\alpha = \beta = -1$. The sole nonsingular radius-eight profile is

$$(\alpha, \beta, h) = (8, -4, 11).$$

Here $j = 3$, and $U = 9 \in \mathcal{U}_{p,11}$; (19) gives

$$(h_U, r_U, s_U, \lambda_U) = \left(1, 3, \frac{p+3}{11}, 1\right),$$

$$a_U = \frac{3(p+3)}{11}, \quad R_U = \frac{p+36}{11}.$$

Thus the earlier radius-eight map is exactly the beta-square specialization of Theorem 9.1.

If a residual state nevertheless has a negative-square coordinate, it obeys the stronger finite exclusions

$$\alpha = -4c^2 \implies \nexists U \mid j^2 \text{ with } U \equiv c^2 \pmod{h},$$

$$\beta = -4c^2 \implies \nexists U \mid j^2 \text{ with } c^2 U \equiv j^2 \pmod{h}.$$

All earlier exact conditions remain: $h \mid \alpha\beta - 1$, $p \equiv \alpha \pmod{h}$, $p\beta \equiv 1 \pmod{h}$, $(h/p) = -1$, neither coordinate is zero or a positive square, and the exterior atlas bound is unchanged.

The overlaps are typed exactly. If one source has both negative-square coordinates, both descriptions parameterize the same full set $\mathcal{U}_{p,h}$; their selected targets coincide exactly when $c_\alpha c_\beta = j$. If δ is the squarefree part of $(R-1)/2$, the diagonal return coincides with a fixed- $Q = h$ target exactly when

$$\delta = h, \quad U = j.$$

Different U 's in $\mathcal{U}_{p,h}$ always produce different original middle states.

9.2. A strict hard-prime improvement. At

$$p = 41161, \quad h = 11, \quad j = 3, \quad c = 5,$$

the original E state

$$(a, R, u, h, r, s, \kappa, D, \alpha, \beta) = (10318, 111, 9678284, 11, 938, 1, 347829, 348767, -100, -348756)$$

has $\alpha = -4 \cdot 5^2$, although $5 \nmid 3$. The state equations are

$$a = 11 \cdot 938, \quad u = 11 \cdot 938^2, \quad 4a - p = 111, \quad 4u + 1 = 111 \cdot 348767,$$

$$pr + s = 111 \cdot 347829.$$

Nevertheless $U = 3 \mid 9$ and $p + 4U = 11 \cdot 3743$, so Theorem 9.1 gives

$$(a_U, R_U, U, h_U, r_U, s_U, \lambda_U) = (11226, 3743, 3, 3, 1, 3742, 1)$$

and

$$\boxed{\frac{4}{41161} = \frac{1}{11226} + \frac{1}{462073386} + \frac{1}{123483}}. \quad (26)$$

Primality is certified by the Lucas witness 22:

$$41160 = 2^3 \cdot 3 \cdot 5 \cdot 7^3, \quad 22^{41160} \equiv 1 \pmod{41161},$$

while the residues at exponents $41160/q$ for $q = 2, 3, 5, 7$ are

$$41160, \quad 40429, \quad 26424, \quad 13707,$$

and each residue minus one is coprime to 41161.

By contrast, at the prime $p = 3049$, the original E state

$$(a, R, u, h, r, s, \kappa, D, \alpha, \beta) = (806, 175, 20956, 31, 26, 1, 453, 479, -144, -448)$$

has ordered denominators $(806, 14043, 1113244782)$. Here $j = 8, c = 6$, and

$$\{U \bmod 31 : U \mid 64\} = \{1, 2, 4, 8, 16\}, \quad c^2 = 36 \equiv 5 \pmod{31}.$$

Hence $\mathcal{U}_{3049,31}$ is empty. Primality is certified by the Lucas witness 11: $11^{3048} \equiv 1 \pmod{3049}$, while the residues at exponents $3048/q$ for $q = 2, 3, 127$ are 3048, 2516, 3041, each with predecessor coprime to 3049. A negative-square coordinate alone does not force a fixed- Q return.

REFERENCES

- [1] Elsholtz, C., & Tao, T. (2013). Counting the number of solutions to the Erdős–Straus equation on unit fractions. *Journal of the Australian Mathematical Society*, 94, 50–105. [arXiv:1107.1010](#), §2, Propositions 2.2 and 2.6.
- [2] Split-Zero programme. (2026). *The ES–Fable inverse correspondence in the original weighted conductor*. Equations GF1–GF11, GF16–GF18 and FC32–FC35. Zeta workbench, revision 369f6ea9e0312bdf348ca9efd294735efc99c51d, source blob f777bb9f1434fa90f2f8bc762860048cc1d32bb4. https://github.com/KokunoYumeto/zeta-function-research-reader/blob/369f6ea9e0312bdf348ca9efd294735efc99c51d/workbenches/splitzero-tandem/continuations/20260920-fable-conductor/FABLE_TO_ORIGINAL_CONDUCTOR.tex.
- [3] The Clankers. (2026). *Turn 7 shape rigidity*, `es_turn7_shape_rigidity_20260919_r1/core.tex`, sections “Original source and the marked cubic” and “Original character constraints”. Preserved predecessor source; its historical verification is not re-certified here.
- [4] Hatcher, A. (2022). *Topology of numbers*. American Mathematical Society. §6.4, quadratic reciprocity and supplementary laws. Author’s online edition: <https://pi.math.cornell.edu/~hatcher/TN/TNbook.pdf>.
- [5] Milne, J. S. (2020). *Algebraic number theory*. Course notes, version 3.08, Chapter 7, “Absolute values; local fields”, Hensel lifting and unramified extensions. <https://www.jmilne.org/math/CourseNotes/ANT.pdf>. Only these standard local-field facts are used; the displayed unit lift is proved and executed directly.

Prime-local reciprocal fibres retain the E/M arithmetic type

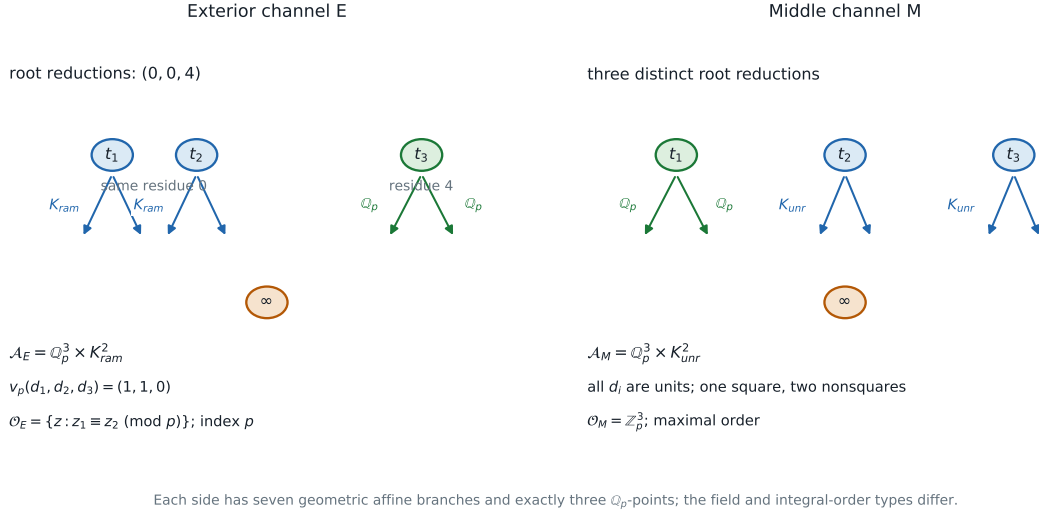


FIGURE 1. The complete prime-local fibre types. Both channels have seven geometric affine branches and three $\mathbb{Q}_p$ -points, but the exterior quadratic factors are ramified while the middle factors are unramified. The display also retains the exact index- p exterior root order. Reproduce with `tools/generate_20260920_illustrations.py`; certify with the local checkers listed in the integration audit.

Complete fixed- Q fibre at $p = 41161, h = 11, j = 3$

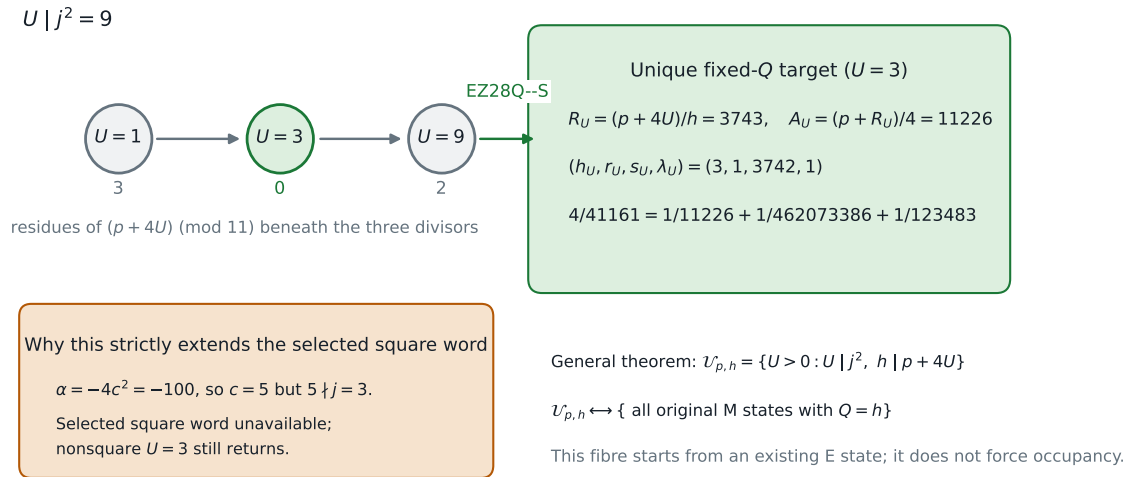


FIGURE 2. The complete fixed- Q divisor fibre in the strict example $p = 41161, h = 11, j = 3$. Of $U = 1, 3, 9$, exactly $U = 3$ satisfies $h \mid p + 4U$, producing the displayed original middle witness even though the earlier selected square word is unavailable. The general bijection and its inverse are proved in the preceding theorem.

- [6] Tao, T. (2026, July 21). *A digestion of the Jacobian conjecture counterexample*. Original three-dimensional Alpöge–Fabel example. <https://terrytao.wordpress.com/2026/07/21/a-digestion-of-the-jacobian-conjecture-counterexample/>. The four-dimensional polynomial here is credited separately to the ES construction as in the Zeta source, not attributed as a fourth original three-dimensional point.